Angles of polygons

ANSWER KEY

Core Concepts: Angles of polygons

1 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 104° ✓

b. 114°

c. 96°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 45^\circ) = 104^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

3 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 13

c. 11 ✓

d. 13

Substituting $x = 1$ into $y = 2x + 9$ gives $y = 2(1) + 9 = 11$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97° ✓

b. 107°

c. 89°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 52^\circ) = 97^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

6 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 11

b. 15

c. 13 ✓

d. 15

Substituting $x = 2$ into $y = 2x + 9$ gives $y = 2(2) + 9 = 13$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 90° ✓

b. 100°

c. 82°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 59^\circ) = 90^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

9 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 13

b. 17

c. 15 ✓

d. 17

Substituting $x = 3$ into $y = 2x + 9$ gives $y = 2(3) + 9 = 15$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying angles of polygons, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 83° ✓

b. 93°

c. 75°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 66^\circ) = 83^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

12 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 15

b. 19

c. 17 ✓

d. 19

Substituting $x = 4$ into $y = 2x + 9$ gives $y = 2(4) + 9 = 17$.

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